

Firewall App

Security Functions for ctrlX OS Version 3

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DOK-XCORE*-FRW***V03**-AP02-EN-P

DC-AE/PAD-SYS (MiSc)/(PiaSt)

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1 About this documentation

Editions of this documentation

Edition	Release date	Note
01	2024-12	First edition Firewall App Version FRW-V-0302
02	2025-07	Edition for Firewall App Version 3.6.0 New content: • ↗ tab – Filter • ↗ tab – “Routing” • ↗ tab – “Directional interfaces” • ↗ tab – Advanced • ↗ Dialog – Add filter rule • ↗ Dialog – Add routing rule • ↗ Dialog – Add interface rule Revised content: • ↗ Firewall App • ↗ Window – Firewall • ↗ Dialog – Add rule

2 Important directions on use

2.1 Intended use

2.1.1 Introduction

Rexroth products are developed and manufactured to the state-of-the-art.

The products are tested prior to delivery to ensure operational safety and reliability.

⚠ WARNING

Personal injury and damage to property due to incorrect use of products!

The products may only be used as intended.

Failure to use the products as intended may cause situations resulting in property damage and personal injury.

NOTICE

Damages resulting from unintended use

Rexroth As the manufacturer does not assume any warranty, liability or compensatory claims for damages resulting from unintended use of the products. The user alone shall bear the risks of an unintended use of the products.

Before using Rexroth products, make sure that all the prerequisites for an intended use of the products are met:

- Personnel that in any way, shape or form uses Rexroth products must first read and understand the relevant safety instructions and be familiar with their intended use
- Leave hardware products in their original state, i.e., do not make any structural modifications. It is not permitted to decompile software products or alter source codes
- Do not install damaged or defective products or commission them
- It has to be ensured that the products have been installed as described in the relevant documentation

2.1.2 Areas of use and application

Products of the ctrlX series are suitable for Motion/Logic applications.

NOTICE

Products of the ctrlX series may only be used with the accessories, mounting parts, and other components specified in this documentation. Components that are not expressly mentioned must neither be attached nor connected. The same applies to cables and lines.

Only to be operated with the hardware component configurations and combinations expressly specified and with the software and firmware specified in the corresponding documentations and functional descriptions.

Products of the ctrlX series are suitable for single-axis as well as for multi-axis drive and control tasks. Device types with different equipment and interfaces are available for using the system in specific applications.

Typical areas of application:

- Building automation
- IoT and Security Gateway or Device
- Handling & Robotic

Controls of the ctrlX CORE series may only be operated under the mounting and installation conditions, in the position of normal use and under the ambient conditions (temperature, degree of protection, humidity, EMC, etc.) specified in the related documentations.

2.2 Unintended use

"Unintended use" refers to using the ctrlX products outside of the above-mentioned areas of application or under operating conditions and technical data other than described and specified in the documentation.

ctrlX products must not be used if they are exposed to following conditions:

- Operating conditions that do not meet the specified ambient conditions. Operation under water, under extreme temperature fluctuations or under extreme maximum temperatures is prohibited
- Applications that have not been expressly authorized by Rexroth

3 Safety instructions

The Safety instructions contained in the available application documentation feature specific signal words (DANGER, WARNING, CAUTION or NOTICE) and, where required, a safety alert symbol (in accordance with ANSI Z535.6-2006).

The signal word is meant to draw the reader's attention to the safety instruction and identifies the hazard severity.

The safety alert symbol (a triangle with an exclamation point), which precedes the signal words DANGER, WARNING and CAUTION, is used to alert the reader to personal injury hazards.

The Safety instructions in this documentation are designed as follows:

 DANGER	In case of non-compliance with this safety instruction, death or serious injury will occur.
 WARNING	In case of non-compliance with this safety instruction, death or serious injury could occur.
 CAUTION	In case of non-compliance with this safety instruction, minor or moderate injury could occur.
NOTICE	In case of non-compliance with this safety instruction, property damage could occur.

4 Introduction and overview

4.1 Firewall App

Function:

The Firewall protects networks or individual computers from unwanted network accesses. Furthermore, they provide functions for network address translation (NAT).

Overview on the network packet filtering (firewall)

The Firewall app uses the Firewall technology Nftables.

Nftables uses tables, chains and rules that are hierarchy levels at the same time. Tables include chains that include the actual Firewall rules.

The Firewall app provides two graphic interfaces of the Nftables functionality. This allows to easily create simple rules ("Basic" license) for package filtering and network address translation (NAT), depending on the license purchased, without thinking about tables and chains.

However, complex rules (require an "Advanced" license) for network traffic/package filtering and network address translation (NAT) can also be created, edited and managed. In the rule editor of the "Advanced" firewall editor, the button  "Add expression" provides a tool for user-friendly creating Firewall rules and rules for NAT using drop-down selection fields, see [Chapter 5.3.4 Dialog – Add rule on page 25](#).

In principle, rules for network packages are created separately according to whether they are intended for the device (Input), for another device and routed through the device (Forward) or output from the device (Output).

Overview on the network address translation (NAT)

Several devices of the internal network can communicate with devices in external networks by using NAT. Devices in an internal network can be addressed via an IP address of external networks.

Source NAT/Masquerade

A packet leaving the device is assigned a different IP address (and port) as sender address than the sending device in the internal network. In the special case Masquerade, it is automatically the IP address of the Ethernet port the device is connected to the external network.

Destination NAT

Packets received by the device are routed to IP addresses and ports of devices of the internal network.

Example Source NAT:

If two devices in the internal network with the same port numbers send a query to the same web server in the external network, not only the sender IP addresses are changed via Source NAT but also the port. The data is stored in the internal NAT table. If the reply of the web server is received, the reply is routed to the correct device in the internal network, using the port stored in the table.

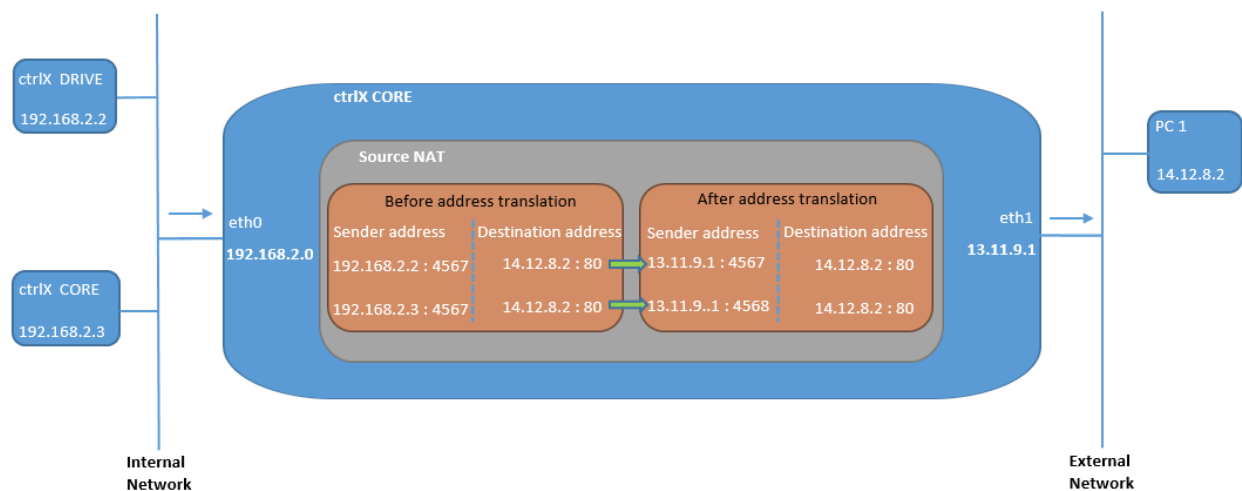


Fig. 1: Example Source NAT

Example Destination NAT:

If devices in the internal network are to be addressed by a device in the external network, addressing is realized using Destination NAT via the IP address of the device which establishes the connection to the external network.

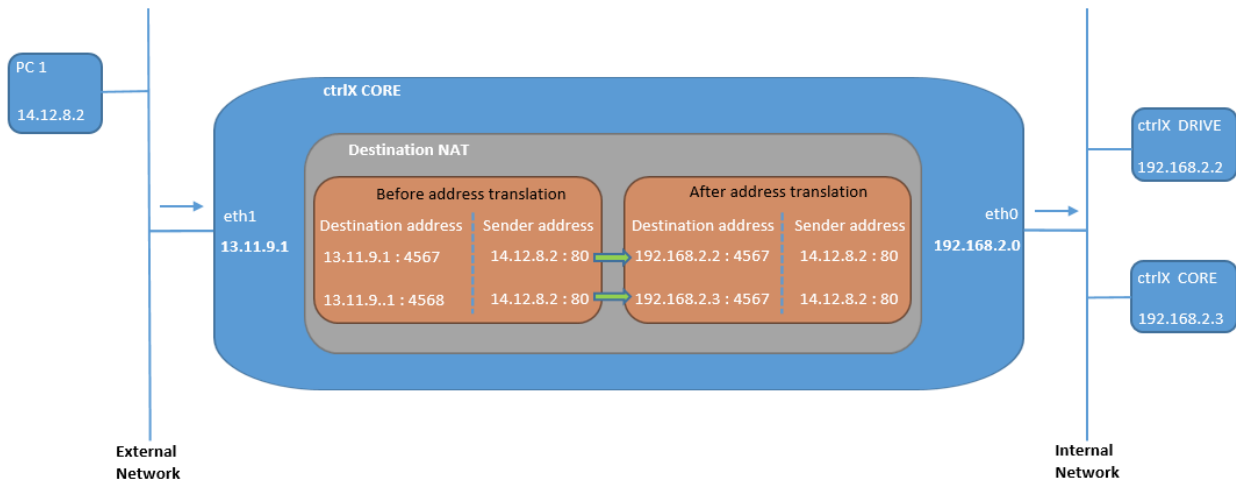


Fig. 2: Example Destination NAT

Further information

- ➔ [Chapter 5.1.1 Window – Firewall on page 8](#)

"Basic" Firewall:

- ➔ [Chapter 5.2.1 tab – Filter on page 9](#)
- ➔ [Chapter 5.2.2 tab – “Routing” on page 11](#)
- ➔ [Chapter 5.2.3 tab – “Directional interfaces” on page 13](#)

"Advanced" Firewall:

- ➔ [Chapter 5.2.4 tab – Advanced on page 14](#)
- ➔ [Chapter 5.3.4 Dialog – Add rule on page 25](#)

5 ctrlX UI – Elements

5.1 Windows

5.1.1 Window – Firewall

In the “Firewall” window, both the "Basic" Firewall as well as the "Advanced" Firewall can be configured. This is performed with the "Basic" license via the three tabs “Filter”, “Routing” and “Directional interfaces” or with the "Advanced" license via the “Advanced” tab.



The configuration of the "Basic" Firewall is visible in the "Advanced" Firewall, but it is read-only and cannot be changed. The rules can be identified by the name that corresponds to a specific convention (_std..._...-inet for IPv4 + IPv6 or _std..._...-ip< x > for IPv4 and IPv6).

The configuration of the "Advanced" Firewall is not visible in the "Basic" Firewall view.

With the "Basic" license, the "Basic" part of the Firewall can be used completely. The "Advanced" part is available, but only read-only, so that the firewall configuration cannot be edited. With the "Advanced" license, the "Basic" part and the "Advanced" part can be used completely.

Site structure

On the start page of the Firewall, there is the “Save firewall configuration” button in the header on the right to make changes persistent. Once the changes are made, but not yet saved, this is indicated to the left of the button with a warning icon “Unsaved changes”.



Changes to the firewall are applied immediately! However, to make changes permanent, the firewall configuration has to be saved explicitly using the “Save firewall configuration” button in the command bar.

If the action “Save firewall configuration” is not executed, all changes made to the Firewall and NAT are lost after restarting the ctrlX OS device.

Below the header on the left, there are the three "Basic" Firewall tabs “Filter”, “Routing” and “Directional interfaces” and the “Advanced” tab on the right to go to the start page of the "Advanced" Firewall.

When opening the Firewall app for the first time, you are in the “Filter” tab of the "Basic" Firewall. Otherwise, the last open tab opens again.

Call:

ctrlX OS side navigation “*Settings → Firewall*”

From here, if only the license for the "Basic" Firewall is available, simple firewall rules can be created via the following tabs.

"Basic" Firewall:

- → Tab “Filter”
- → Tab “Routing”
- → Tab “Directional interfaces”

If an "Advanced" Firewall license is available, complex firewall rules can be created via the following tab:

"Advanced" Firewall:

- → “Advanced” tab

Further information

- ➔ [Chapter 4.1 Firewall App on page 6](#)
- ➔ [Chapter 5.2.1 tab – Filter on page 9](#)
- ➔ [Chapter 5.2.2 tab – “Routing” on page 11](#)
- ➔ [Chapter 5.2.3 tab – “Directional interfaces” on page 13](#)
- ➔ [Chapter 5.2.4 tab – Advanced on page 14](#)
- ➔ [Chapter 5.3.1 Dialog – Add filter rule on page 19](#)
- ➔ [Chapter 5.3.2 Dialog – Add routing rule on page 21](#)
- ➔ [Chapter 5.3.3 Dialog – Add interface rule on page 24](#)
- "ctrlX OS - Runtime", Application Manual (R911423381), chapter ➔ [Licenses - Overview](#)
- "ctrlX OS Runtime", Application Manual (R911423382), chapter ➔ [Configuration - Introduction](#)

5.2 Tabs

5.2.1 tab – Filter

tab of the "Basic" Firewall with the table of the created network filter rules and the “Add filter rule” dialog to be opened using the button and the field to set the default action for data packages that do not match the rules.

Call:

ctrlX OS side navigation “Settings → Firewall” → Tab “Filter”

Overview

If no filter rules have been created yet, there is no command bar and no table on the tab, but the note “No filter rules available” and below the button + “Add filter rule” and an info box with information on the function of filter rules.

If filter rules have already been created, there is a command bar at the top of the tab in which the number of existing elements in the table below is displayed on the left and the drop-down selection field “Default action for non-matching data packets

” which applies to all rules in the table. On the far right is the  “Add filter rule” button. The table showing the filter rules and their characteristics is located below the command bar.

The order of the rules in the table corresponds to the processing sequence. The rules can be shifted in the table using drag-and-drop.

Default action “Drop” for filter rules

If the filter rules are not configured correctly, the system may no longer be accessible after selecting Drop as default action for data packages that do not match the rules!

After the first rule in the table for which the data package does not match the settings in the rule, the data package is dropped. The other rules in the table are no longer applied to the data package.

**Elements of the “Filter” tab**

GUI element	Description
Command bar	“[x] items” Number of filter rules in the table
	Drop-down selection field “Default action for non-matching data packets” • “Accept” – Data packages are accepted • “Drop” – Data packages are dropped without informing the source
	 button Opens the “Add filter rule” dialog
Table	“Description” Entered description of the filter rule
	“Direction” Direction of the network packages to which the rule applies (Input, Output, Forward)
	“Protocol” IP version and protocol type to which the rule applies. (IPv4/IPv6 TCP, UDP, * = any)
	“Source adress” Source IP address of the network package (no entry, for “Direction” Output)

GUI element	Description
	"Port" Source port of the network package (no entry, if "Protocol" = * (any))
	"Destination adress" Target IP address of the network package (no entry, for "Direction" Input)
	"Port" Target port of the network package (no entry, if "Protocol" = * (any))
	"Action on match" Action ("Accept", "Reject", "Drop") executed for network packages that match all criteria of the rule
	"Enabled" Status display and slider for the state of the filter rule (enabled/disabled)
	"Actions"  Opens the dialog for the filter rule
	 Filter rule deletion after confirming a security prompt

Further information

- ➔ [Chapter 4.1 Firewall App on page 6](#)
- ➔ [Chapter 5.1.1 Window - Firewall on page 8](#)
- ➔ [Chapter 5.3.1 Dialog - Add filter rule on page 19](#)
- ➔ [Chapter 5.3.2 Dialog - Add routing rule on page 21](#)
- ➔ [Chapter 5.3.3 Dialog - Add interface rule on page 24](#)
- ➔ [Chapter 5.3.4 Dialog - Add rule on page 25](#)
- ➔ [ctrlX OS Runtime, Application Manual, chapter "Window - "Manage app data"](#)

5.2.2 tab – “Routing”

"Basic" Firewall tab with the table of created routing rules and the dialog "Add routing rule" to be opened via the  button

Call:

ctrlX OS side navigation "Settings → Firewall" → Tab "Routing"

Overview

If no routing rules have been created yet, there is no command bar and no table on the tab, but the note "No routing rules available" and below the button + "Add routing rule" and an info box with information on the function of routing rules

If routing rules have already been created, there is a command bar at the top of the tab in which the number of existing elements in the table below is displayed on the left and the  "Add routing rule". The table showing the routing rules and their characteristics is located below the command bar.



The order of the rules in the table corresponds to the processing sequence. The rules can be shifted in the table using drag-and-drop

Elements of the “Routing” tab

GUI element	Description
Command bar	“[x] items” Number of routing rules in the table
	button Opens the “Add routing rule” dialog
Table	“Description” Entered description of the routing rule
	“Protocol” IP version and protocol type to which the rule applies. (IPv4/IPv6 TCP, UDP, * = any)
	“Source address” Source IP address of the network package
	“Port” Source port of the network package. (Entry * = any. No entry, for “Protocol” = * (any))
	“Destination address” Target IP address of the network package
	“Port” Target port of the network package (no entry, if “Protocol” = * (any))
	“NAT-Type” Source NAT, Masquerade or Destination NAT
	“Translate to” No entry for NAT type Masquerade. For NAT type Source NAT or Destination NAT IP address and optional port
	“Enabled” Status display and slider for the state of the routing rule (enabled/disabled)
	“Actions” Opens the dialog for the routing rule Routing rule deletion after confirming a security prompt

Further information

- ➔ Chapter 4.1 Firewall App on page 6
- ➔ Chapter 5.1.1 Window – Firewall on page 8
- ➔ Chapter 5.3.1 Dialog – Add filter rule on page 19
- ➔ Chapter 5.3.2 Dialog – Add routing rule on page 21
- ➔ Chapter 5.3.3 Dialog – Add interface rule on page 24

- ➔ Chapter 5.3.4 Dialog – Add rule on page 25
- ➔ ctrlX OS Runtime, Application Manual, chapter "Window - "Manage app data"

5.2.3 tab – “Directional interfaces”

Tab of the "Basic" Firewall with the table of the created interface rules and the dialog “Add interface rule” to be opened using the  button.

Call:

ctrlX OS side navigation “Settings → Firewall” → tab “Directional interfaces”

Overview

If no interface rules have been created yet, there is no Command bar and no table but the note “No interface rules available” and below the button + “ Add interface rule” and an info box with information on the function of interface rules

If interface rules have already been created, there is a command bar at the top of the tab in which the number of existing elements in the table below is displayed on the left and the  button is located on the right Add interface rule”. The table showing the interface rules and their characteristics is located below the command bar.



The order of the rules in the table corresponds to the processing sequence. The rules can be shifted in the table using drag-and-drop

Elements of the “Directional interfaces” tab

GUI element	Description
Command bar	“[x] items” Number of interface rules in the table
	 button Opens the “Add interface rule” dialog
Table	“Description” Entered description of the interface rule
	“Interface” Device interface for which the rule applies
	“Incoming traffic” The rule set for the interface for data package that reach the device
	“Outgoing traffic” The rule set for the interface for data package that leave the device
	“Prevent lockout” Status of the option of the engineering interface
	“Enabled” Status display and slider for the state of the interface rule (enabled/disabled)

GUI element	Description
	“Actions”  Opens the dialog for the interface rule  Interface rule deletion after confirming a security prompt

Further information

- ➔ [Chapter 4.1 Firewall App on page 6](#)
- ➔ [Chapter 5.1.1 Window – Firewall on page 8](#)
- ➔ [Chapter 5.3.1 Dialog – Add filter rule on page 19](#)
- ➔ [Chapter 5.3.2 Dialog – Add routing rule on page 21](#)
- ➔ [Chapter 5.3.3 Dialog – Add interface rule on page 24](#)
- ➔ [ctrlX OS Runtime, Application Manual, chapter "Window - "Manage app data"](#)

5.2.4 tab – Advanced

To create and edit rules in the “Advanced” firewall tab, an "Advanced" firewall license is required. In contrast to the "Basic" Firewall, it is possible to create complex rules.

At the top of the tab is a schematic overview on the packet processing of the firewall, for the various filter rule chain hooks “Input filter”, “Forward filter” and “Output filter” are located below. For the list of available chains in tabular form, go below the tabs of the tables “IPv4 + IPv6”, “IPv4” and “IPv6”. The chains are listed with “Priority”, “Name” and “Default policy”. In the “Actions” column, open the chain editor and delete the chain using the buttons “Edit” and “Delete”.

On the sites of the tiles “Destination NAT, ” and “Source NAT”, the list of available chains in tabular form is shown below the tabs of the tables “IPv4” and “IPv6”. The chains are listed with “Priority” and “Default policy”. In the “Actions” column, open the chain editor and delete the chain using the buttons “Edit” and “Delete”.

In the rule editor, the button  “Add expression” provides a tool for user-friendly creation of firewall rules and rules for NAT using drop-down selection fields, see ➔ [Chapter 5.3.4 Dialog – Add rule on page 25](#)



The configuration of the "Basic" Firewall is visible in the "Advanced" Firewall, but it is read-only and cannot be changed. The rules can be identified by the name that corresponds to a specific convention (_std..._...-inet for IPv4 + IPv6 or _std..._...-ip< x > for IPv4 and IPv6).

The configuration of the "Advanced" Firewall is not visible in the "Basic" Firewall view.

With the "Basic" license, the "Basic" part of the Firewall can be used completely. The "Advanced" part is available, but only read-only, so that the firewall configuration cannot be edited. With the "Advanced" license, the "Basic" part and the "Advanced" part can be used completely.

Call:

ctrlX OS side navigation “Settings → Firewall” → Tab “Advanced”

Further information

- ➔ [Chapter 4.1 Firewall App on page 6](#)
- ➔ [Chapter 5.1.1 Window – Firewall on page 8](#)
- ➔ [Chapter 5.3.1 Dialog – Add filter rule on page 19](#)
- ➔ [Chapter 5.3.2 Dialog – Add routing rule on page 21](#)
- ➔ [Chapter 5.3.3 Dialog – Add interface rule on page 24](#)
- ➔ [Chapter 5.3.4 Dialog – Add rule on page 25](#)
- "ctrlX OS Runtime", Application Manual (R911423382), chapter
➔ [Configuration - Introduction](#)

Elements of the Firewall

Tables

There are the following tables in the Firewall app:

- **IPv4:**
Table of the IPv4 family views the IPv4 network traffic and includes the respective rules
- **IPv6:**
Table of the IPv6 family views the IPv6 network traffic and includes the respective rules
- **IPv4 + IPv6 (not for NAT):**
The IPv4 + IPv6 family table views both the IPv4 and the IPv6 network traffic and contains rules for both. Rules for IPv4 packets do not affect IPv6 packets. Rules for both protocols affect the packets of both

Chains

Chains contain rules for the tables IPv4 + IPv6 (not for NAT), IPv4, IPv6 of the following network packet filter chain hooks and network address translation (NAT):

- **Input filter**
The rules in these chains affect the network packets received by the device
- **Forward filter**
The rules in these chains affect the network packets passing the device
- **Output filter**
The rules in these chains affect the network packets leaving the device
- **Destination NAT**
The rules in these chains affect the network packets received by the device
- **Source NAT**
The rules in these chains concern network packets leaving the device

Rules for network packet filtering

Rules consist of expressions and statements which chain is to be inserted. Rules for network packets filtering have the effect that one of the actions "Accept", "Drop" or "Reject" is applied to the network packets.

If a packet is accepted by a chain and if there is another chain with the same fulfilled expression and a lower priority, the packet subsequently runs through this other chain. An acceptance decision for this packet by a rule or by the Default policy is thus only finalized if the last run through chain also triggers the "Accept" action. However, the same does not apply to packets assigned by a rule or the "Default policy" with a "Drop" action. Instead, discarding of packets becomes active immediately without evaluating more rules or chains.

Rules for network address translation (NAT)

For “Source NAT” the expressions, which are usually defined, cause packets leaving the device to be assigned with a different IP address and port than the sending device. In the special case “Masquerade”, this is automatically the IP address of the Ethernet port connected to the external network. For “Destination NAT”, the expressions of the rule have the effect that affected network packets received by the device on the externally visible address, are routed to other internal IP addresses and ports of downstream devices. The Accept, Drop or Reject actions do not apply to NAT rules.

Examples of Firewall/NAT rule chain structure:

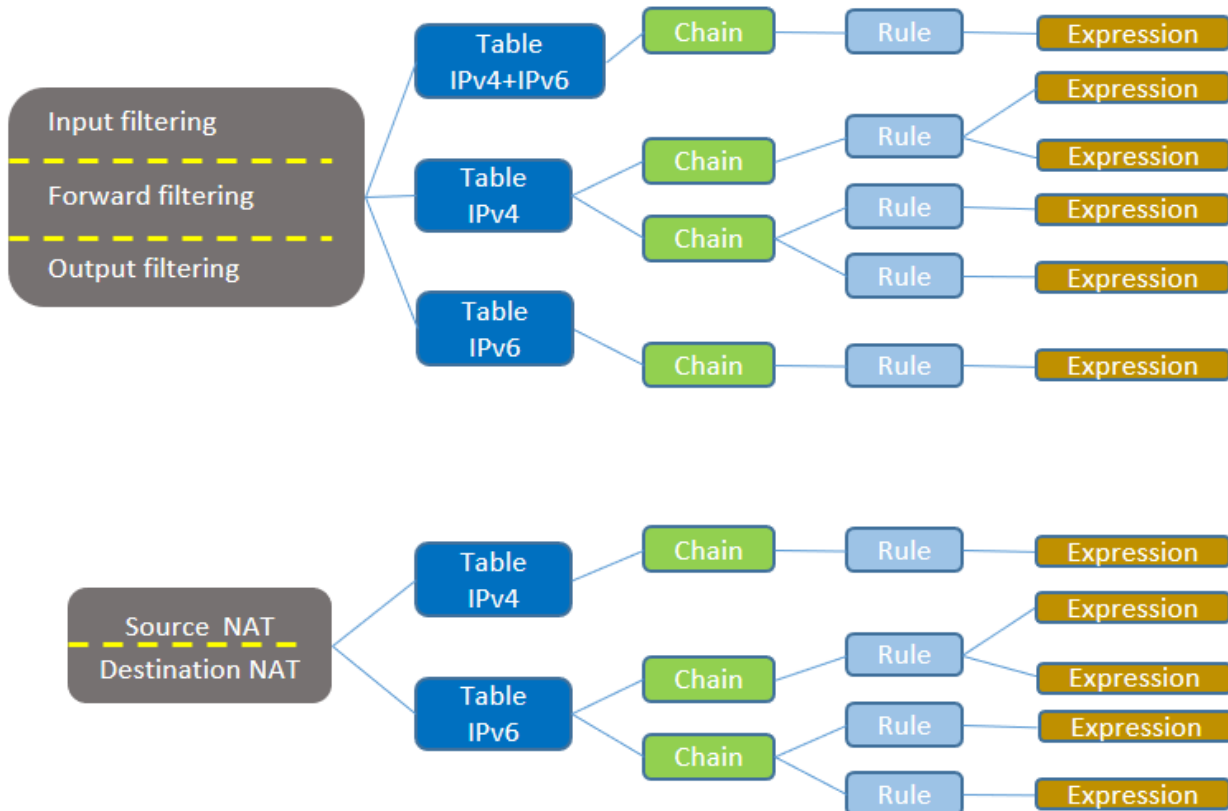


Fig. 3: Elements of the Firewall

Chains

Overview on network packet filtering

Chains contain the actual Firewall rules that are applied to the network packets detected by the specific chain. For packets without any applied rules, the Default policy of the chain applies. The default setting of the Default policy of chains is “Accept”. The priority of a chain specifies the processing sequence. Lower values have a higher priority. For example: -100 has a higher priority than -50 and 20 has a higher priority than 30. The default setting of the chain Priority for network packet filtering is 0.

Overview on the network address translation

These chains contain the rules for Destination NAT/Source NAT. These chains do not have a “Default policy”. The “Priority” of a chain specifies the processing sequence. Lower values have a higher priority. The default setting of the chain Priority for Destination NAT is higher than for Source NAT chains.

Create

A rule chain for a table “IPv4 + IPv6” (*not for NAT*), “IPv4” or “IPv6” for one of the entry points “Input filter”, “Forward filter”, “Output filter” and “Destination NAT” and “Source NAT” is created on the respective configuration page via the button  “Add chain” in the command bar of the table.

A “Name” has to be specified in the “Add chain” creation dialog.

Click on the “Add chain” button, the chain is created and displayed in the list with “Priority”, “Name” and the default “Default policy”.

Creating a chain with the Default policy“Drop”

To configure a chain with the “Default policy”“Drop”, create it with the “Default policy” setting “Accept” first. Before configuring your rules, create rules to accept packets of the web browser of the Engineering PC to ctrlX OS device and vice versa.

The rules for Localhost are required for internal communication purposes.

Create the following expressions for “Input filter” in the "Rule(s)" chain:

Firewall > Input filter >

DefaultChain (IPv4+IPv6)

  boschrexroth  

Name

DefaultChain

Priority

0

Default policy

drop

▼

Default policy is "drop". Rules should accept matching packets.

Rules

3 items



Name	Expressions	Action if match	Actions
Allow https	match payload: tcp dport == 443	accept	 
Allow http	match payload: tcp dport == 80	accept	 
Allow localhost	match payload: ip saddr == 127.0.0.1 match payload: ip daddr == 127.0.0.1	accept	 

To reorder rules drag & drop the corresponding rows

Fig. 4: Input filter, required rules for chains with the Default policy "Drop"

Create the following expressions for “Output filter” in the "Rule(s)" chain:

Firewall > Output filter > DefaultChain (IPv4+IPv6)

Priority 0

Default policy drop

Default policy is "drop". Rules should accept matching packets.

Rules

2 items

Name	Expressions	Action if match	Actions
Allow answer	match ct: direction == reply	accept	
Allow localhost	match payload: ip saddr == 127.0.0.1 match payload: ip daddr == 127.0.0.1	accept	

To reorder rules drag & drop the corresponding rows

Fig. 5: Output filter, required rules for chains with the Default policy "Drop"

Save the configuration using the "Firewall configuration" button on the main Firewall homepage. Once you have ensured that you can still access the web interface of the ctrlX OS device's web interface, you can switch the Default policy to "Drop".



Please note:

For more services, more rules can be required with the action "Accept".

Editing

Select the icon to open the editor for the chains on the site of the tile of the respective chain hook for network packet filters or NAT in the table of the existing chains.

The navigation bar is in the header of the editor and can be used to jump back within the structure.

Fields to edit "Name", "Priority" and "Default policy" (not applicable to NAT) of the chain are located below the window name.

The rules of the chain are listed below in the table "Rules" with "Name", "Expressions" and "Action if match" (not applicable to NAT). In the command bar above the table, there are the buttons (on the right) "Add rule" to create a new rule and "Watch counters/Stop watching counters" if there is at least one counter in the rule. In the "Actions" column, there are buttons to open the rule editor and to delete the rule. There are no rules yet in the newly created and default chains.

Using drag-and-drop, a rule can be moved up and down in the list, thus changing the application order of the rules in the chain.



Applying changes to the ctrlX OS device

Changes to chains and in the rules contained in the chains are immediately applied to the ctrlX OS device if they are applied to the control editor in the chain with

To ensure that the changes are also applied after restarting the ctrlX OS device, click on the "Save Firewall configuration" button in the Firewall command bar.

The configuration of the Firewall is stored in the active configuration of the ctrlX OS device. If the archive of a configuration is loaded using "Manage app data", the Firewall configuration overwrites the Firewall configuration currently active in the ctrlX OS device.

Rules

Overview on the network packet filtering

Rules consist of individual expressions for “Match packet payload”, “Match packet meta data” or “Match packet conntrack data”. Their position in the rule determines the sequence to be applied to the network packets and the position of the rule in the chain.

Overview on the network address translation (NAT)

The NAT rules always contain at least two expressions. A transport protocol expression on the first position on “Match packet payload”, “Match packet meta data” or “Match packet conntrack data” and the expression for “Source NAT”/“Masquerade” or “Destination NAT” in the last position.

Overview on expressions

Expressions within a rule are evaluated consecutively. If the first expression is true, the next expression is evaluated, etc. If **all** expressions in a rule are true, the specified action is executed.

Example

When creating a rule with the “action on match” **Reject** that contains the expressions **match payload: tcp sport ==50** and **match payload: tcp sport ==60**, the action “Reject” of the rule is not executed as the source port(“sport”) cannot be 50 and 60 at the same time.

Create

Multiple chains can be created in the tables, “IPv4 + IPv6 ” (*not applicable to NAT*), “IPv4” and “IPv6” of the chain hooks “Input filter”, “Output filter” and “Forward filter” as well as “Destination NAT” and “Source NAT”. For each chain, several rules can be created, each of which contains expressions that trigger the Accept, Drop or Reject action.

For NAT, there is no Accept, Drop or Reject action available. If the rule is complied with, the sender IP address and optionally the port is changed in the header of the network packets for “Source NAT”/“Masquerade”. For “Destination NAT”, the target IP address and optionally the port is changed.

To create a rule, the Firewall app provides an editor opened from each chain via the “Add rule”  button in the Command bar of the “Rules” table, see [Chapter 5.3.4 Dialog – Add rule on page 25](#).

If no rule was created for a chain, no Command bar and no table but only the button + “Add rule” is displayed on the chain site.

After adding a rule to the button, command bars and tables are displayed on the chain site with the rule entry.

5.3 Dialogs

5.3.1 Dialog – Add filter rule

Dialog of the “Basic” Firewall to create a network filter rule. By selecting the characteristics for the filter rule (IP addresses, protocol, ports), specific rules are created for data packages that are intended for the device (Input), are routed through the device (Forward) or are sent from the device (Output). Due to the set “Action on match” of the filter rule, if a package matches all the criteria of the rule, the package is then accepted (“Accept”) or rejected (“Reject”) or dropped (discarded) (“Drop”).

Call:

ctrlX OS side navigation “Settings → Firewall ” → Tab “Filter” →  Add filter rule

Elements of the “Add filter rule” dialog

Element		Description
Option: “Enabled”		The filter rule is active when the slider is on the right and highlighted in blue
Input field: “Description”		Text field to optionally provide a description for the rule
“Matching criteria:”	Drop-down selection field: “Direction”	• Forward Affects packages routed through this device • Input Affects packages addressing this device. The Destination address of the package is the same as the IP address of this device • Output Affects packages originating from this device. The Source address of the package is the same as the IP address of this device
	Drop-down selection field: “IP version”	IPv4 or IPv6
	Drop-down selection field: “Protocol”	TCP, UDP, * (any)
	Input field: “Source address*” (Not available for “Direction” = Output)	• Any address: * • Or enter a single address, an address range or a subnet
	Input field: “Source port*” (Not available for “Protocol” = * (any))	• Any port: * • Or enter a single port, a port list, a port range or a combination
	Input field: “Destination address*” (Not available for “Direction” = Input)	• Any address: * • Or enter a single address, an address range or a subnet

Element		Description
	Input field: “Destination port*” (Not available for “Protocol” = * (any))	- Any port: * - Or enter a single port, a port list, a port range or a combination
“Action on match”	Drop-down selection field: “Action”	“Accept” The network packet is accepted if it matches the matching criteria. “Reject” If the network packet matches the matching criteria, it is rejected and the sender is informed. “Drop” If the network packet matches the matching criteria, it is dropped without informing the sender
button		The filter rule is applied and entered into the table in the “Filter” tab
		A query dialog opens where to cancel the closing of the dialog via the  button and to continue editing. Alternatively, the settings made can be saved using the  button or dropped using the  button and the dialog is closed

Further information

- ➔ Chapter 5.2.1 tab – Filter on page 9
- ➔ Chapter 5.1.1 Window – Firewall on page 8
- ➔ Chapter 4.1 Firewall App on page 6
- ➔ Chapter 5.1.1 Window – Firewall on page 8
- ➔ ctrlX OS Runtime, Application Manual, chapter "Windows - Manage app data"

5.3.2 Dialog – Add routing rule

Dialog of the "Basic" Firewall to create a routing rule. Specific rules for the IP address and port compilation are created by selecting the characteristics for the routing rule.

This can be used to replace source IP address and port for outgoing network package. “(NAT type”: Source NAT).

“NAT type” Masquerade replaces the source IP address with the current IP address of the outgoing interface.

Target IP address and port can be replaced for incoming network package“(NAT type”: Destination NAT).

Call:

ctrlX OS side navigation “Settings → Firewall” → Tab Routing →  Add routing rule

Elements of the “Add routing rule” dialog

GUI element		Description
Option “Enabled”		The routing rule is active when the slider is on the right and highlighted in blue. The default setting is “Enabled”
Input field: “Description”		Text field to optionally provide a description for the rule
“Matching criteria”:	Drop-down selection field: “IP version”	IPv4 or IPv6
	Drop-down selection field: “Protocol”	TCP, UDP, * (any)
	Input field: “Source address*”	<i>Original source IP address of the data package</i> - Any address: * - Or enter a single address, an address range or a subnet
	Input field: “Source port*” (Not available for “Protocol” = * (any))	Original source port of the data package - Any port: * - Or enter a single port, a port list, a port range or a combination
	Input field: “Destination address*”	<i>Original target IP address</i> - Any address: * - Or enter a single address, an address range or a subnet
	Input field: “Destination port*” (Not available for “Protocol” = * (any))	<i>Original target port</i> - Any port: * - Or enter a single port, a port list, a port range or a combination

GUI element		Description
"Address translation (NAT) on match"	Drop-down selection field: "NAT type"	- Source NAT Replaces the source IP address and optionally the source port of outgoing network packages with the specified address and port. - Masquerade Replaces the source IP address of outgoing network packages with the current IP address of the outgoing interface. - Destination NAT Replaces the target IP address and optionally the target port of incoming network packages with the specified address and port
	Input field: "Address to translate to*" (Not available for "NAT type" = Masquerade)	IPv4 or IPv6 address to be changed to depends on the set IP version
	Input field: "Port to translate to*" (Not available for "NAT type" = Masquerade)	Port to be changed to. The specification of a port is optional
button		The routine rule is applied and entered into the table in the "Routing" tab
		A query dialog opens where to cancel the closing of the dialog via the  button and to continue editing. Alternatively, the settings made can be saved using the  button or dropped using the  button and the dialog is closed

Further information

- ➔ Chapter 5.2.2 tab – "Routing" on page 11
- ➔ Chapter 5.1.1 Window – Firewall on page 8
- ➔ Chapter 4.1 Firewall App on page 6

- ➔ Chapter 5.1.1 Window – Firewall on page 8
- ➔ ctrlX OS Runtime, Application Manual, chapter "Windows - Manage app data"

5.3.3 Dialog – Add interface rule

Dialog of the "Basic" Firewall to create an interface rule. By selecting simple options for the interfaces of a ctrlX OS device, separate according to incoming and outgoing data traffic, rules can be created for network packets that reach or leave the device via the selected interface.

Call:

ctrlX OS side navigation “*Settings* → *Firewall*” → Tab Directional interfaces →  Add interface rule

Elements of the “Add interface rule” dialog

Element	Description
Option: “Enabled”	The interface rule is active when the slider is on the right and highlighted in blue
Input field: “Description”	Text field to optionally provide a description for the interface rule
Drop-down selection field “Interface name*”	Interface for which the rule is created (number of listed interfaces depends on the hardware)
Option “Prevent user interface lockout”	Enable this option for interfaces used for web-based configuration to ensure that the configuration settings made do not lock yourself out of the device
“Incoming traffic”	• “Accept all” All network packages via the set interface are accepted. • “Reject all except established connections” Network packets of set up connections via the set interface are accepted, network packets of new connections are blocked and the clients are informed. • “Drop all except established connections” Network packets of set up connections via the set interface are accepted, network packets of new connections are blocked without informing the clients. • “Reject all” Blocks all network packets via the set interface and informs clients. • “Drop all” Blocks all network packets via the set interface without informing clients

Element	Description
“Outgoing traffic”	• “Accept all” All network packages via the set interface are accepted. • “Reject all except established connections” Network packets of set up connections via the set interface are accepted, network packets of new connections are blocked and the clients are informed. • “Reject all” Blocks all network packages via the set interface without informing clients
button 	The interface rule is applied and entered into the table in the “Directional interfaces” tab
button 	A query dialog opens to cancel the closing of the dialog via the  button and to continue editing. Alternatively, the settings made can be saved using the  button or dropped using the  button and the dialog is closed

Further information

- ➔ [Chapter 5.2.3 tab – “Directional interfaces” on page 13](#)
- ➔ [Chapter 5.1.1 Window – Firewall on page 8](#)
- ➔ [Chapter 4.1 Firewall App on page 6](#)
- ➔ [Chapter 5.1.1 Window – Firewall on page 8](#)
- ➔ [ctrlX OS Runtime, Application Manual, chapter "Windows - Manage app data"](#)

5.3.4 Dialog – Add rule

Dialog of the "Advanced" Firewall to create a rule in a chain using the editor to insert expressions.

Call:

ctrlX OS side navigation “Settings → Firewall → Selection element (e.g. Input filter) → Table selection (e.g. IPv4) → Chain selection” →  "Edit" →  "Add rule"

Overview

Rules consist of individual expressions. Their respective position in the rule determines the sequence used to apply the expressions to the network packets and the position of the rule in the chain.

Expressions within a rule are evaluated consecutively. If the first expression matches, the next expression is evaluated, and so on. If **all** expressions of a rule are true, the specified rule is executed.

Example

When creating a rule with the “action on match” **Reject** that contains the expressions **match payload: tcp sport ==50** and **match payload: tcp sport ==60**, the action “Reject” of the rule is not executed as the source port(“sport”) cannot be 50 and 60 at the same time.

Creating a rule for network packet filtering

Multiple rules can be created for the chains in the tables “IPv4 + IPv6”, “IPv4” and “IPv6” of the entry points “Input filter”, “Output filter” and “Forward filter”. They consist of one or multiple expressions. A rule triggers an action “Accept”, “Drop” or “Reject”. Open the editor of the chain to create a rule for the chain. Select the **+** “Add Rule” button in the editor of the chain to open the editor for the chain rules.

Creating a rule for network address translation

Multiple rules can be created for the chains in the tables “IPv4” and “IPv6” for “Destination NAT” and “Source NAT”. They consist of one or multiple expressions.

For NAT, there is no Accept, Drop or Reject action available triggering the rule. If the rule is applied, the sender IP address and optionally the port is changed in the header of the network packet for Source NAT/Masquerade. For Destination NAT, the target IP address and optionally the port is changed.

Open the editor of the chain to create a rule for the chain. Select the **+** “Add rule” button in the editor of the chain to open the editor for the chain rules.

Elements of the “Rule” editor

Specify a name for the rule in the “Name” field. A comment to the rule can be entered into the “Comment” field. Find the “Add expression” **+** button to the right in the header of the “Expressions” list. Use this button to add different expressions that are part of a rule to the rule.

Elements for network packet filtering

Click on the **+** “Add expression” button to open a drop-down box with the different groups of network packet elements and a contour element

The screenshot shows the 'Rule' editor interface. At the top, there's a 'Rule' title and a close button (X). Below it are two input fields: 'Name *' with the placeholder 'New rule' and 'Comment'. Underneath is the 'Expressions' section, which contains a table with columns 'Type', 'Condition', and 'Actions'. The table is currently empty, with a message 'No expressions available.' and a prompt 'To add an expression, click +'. To the right of the table, there's a blue button with a white '+' icon. A dropdown menu is open from this button, listing four options: 'Match packet payload', 'Match packet meta data', 'Match conntrack data', and 'Counter'.

Fig. 6: Rule editor for network packet filtering

The element “Counter” is directly inserted in the rule when clicking on the button. Click on another entry to open the editor. An expression can be configured with elements from the selected group “Match packet payload”, “Match packet meta data”, “Match conntrack data”. Depending on the value selected in the expression editor in the field to the left, the input option/value selection changes in the field to the right. To enter the value, the rule expression verifies, the option is provided in the editor, where practical, to enter a value range for the element group “Match packet payload” using the option “Specify range”.

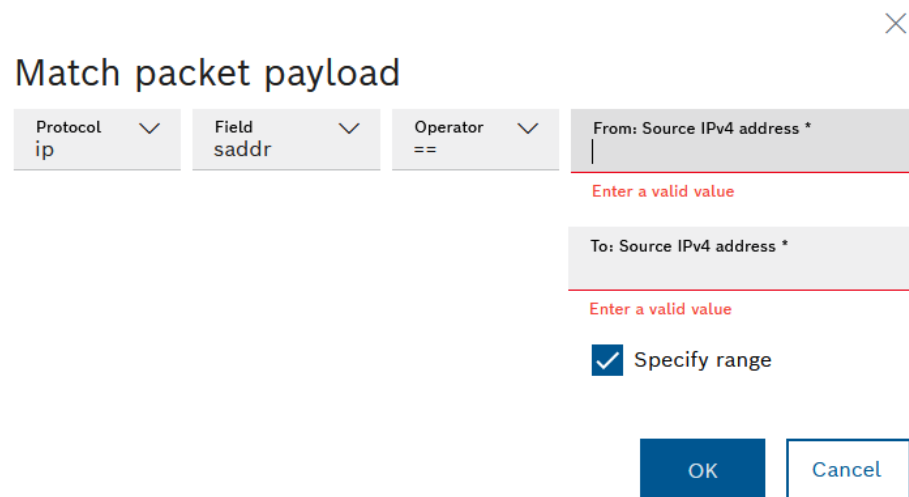


Fig. 7: Specify example for range

If there are valid entries in all fields, the expression is transferred to the “Expressions” list by clicking on the  button.

The expressions can be moved within the rule via drag-and-drop, which changes their processing within the rule. The buttons in the column “Actions” of the “Expressions” list are used to open the editor for the respective expression or to delete the expression from the rule. In the last line of the editor, in the field “Action if Match”, select one of the actions “Accept”, “Drop” or “Reject” to be triggered by the rule. Click on OK  to add the rule of the chain or discard it with Cancel .

Elements for network address translation

Click on the  “Add expression” button in the NAT rule editor to open a drop-down box with the different groups of network packet elements and a contour element as well as elements for “Source NAT” and “Masquerade” or “Destination NAT”.

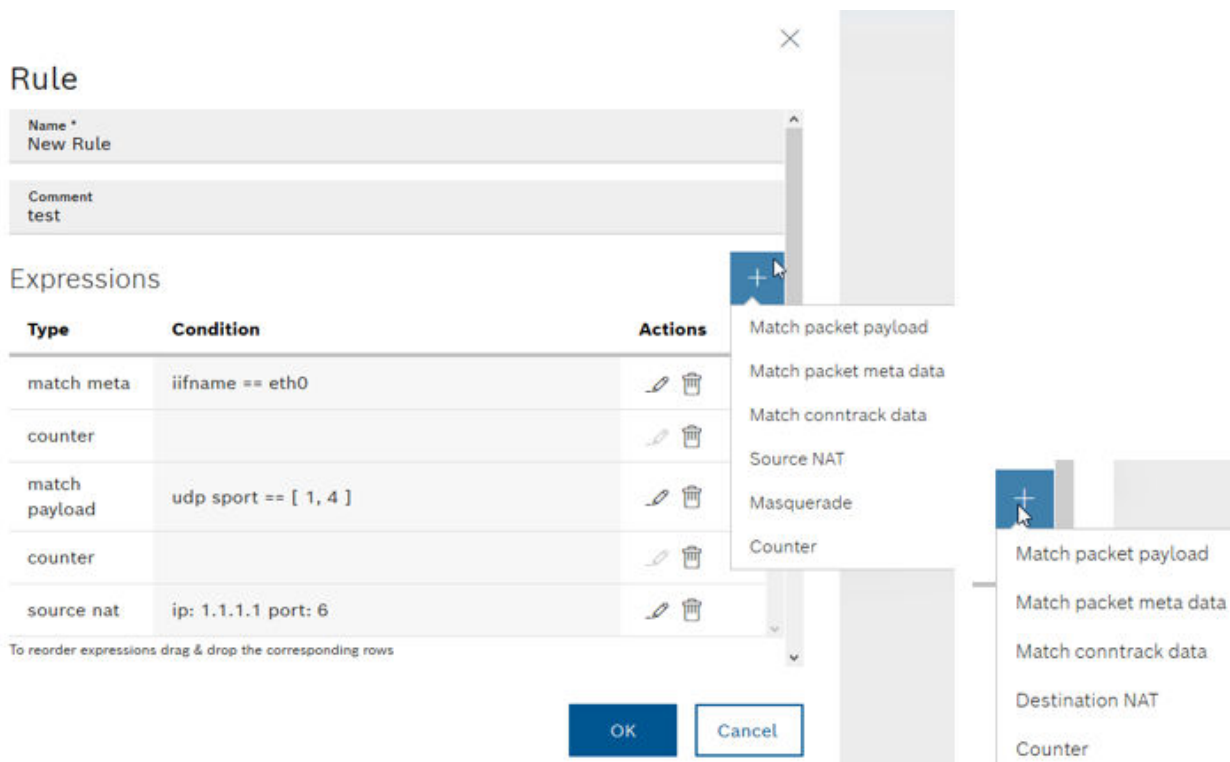


Fig. 8: Example NAT control editor

Click on one of the entries “Source NAT”, “Masquerade” or “ Destination NAT” to open the editor. An expression can be configured using elements of the selected group.

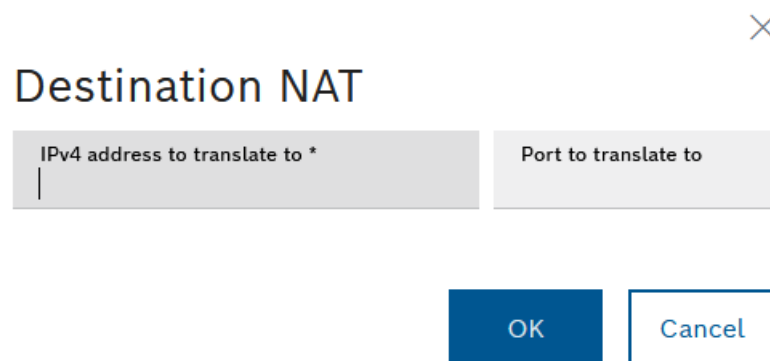


Fig. 9: Example Add expression editor for NAT

If there are valid entries in all fields, the expression is transferred to the “Expressions” list by clicking on the **OK** button.

The expressions can be moved within the rule via drag-and-drop, which changes their processing within the rule. However, the expression for “Source NAT” or “Masquerade” or “Destination NAT” always has to be at the last position. The buttons in the “Actions” column of the list “Expressions” open the editor for the respective expression or delete the expression from the rule. To add the rule to the chain, click on the **OK** button. To discard it, click on the **Cancel** button.

Editing

To edit and delete a rule for a chain in one of the tables “IPv4 + IPv6” (not applicable to NAT), “IPv4” and “IPv6” of the chain hooks “Input filter”, “Output filter”, “Forward filter” as well as “Destination NAT” and “Source NAT”, click on the respective icons in the “Actions” column of the table.

Using drag-and-drop, a rule can be moved up and down within the chain, and thus in the processing sequence.

The editor opened via the  button is the same editor as for the creation of a rule.



Applying changes to the ctrlX OS device

Changes to rules and the contained expressions take effect immediately on the ctrlX OS device when they are applied to the rule by clicking on the  button in the rule editor.

To ensure that the changes also take effect after a restart of the ctrlX OS device, they have to be saved with the button “Save Firewall configuration” on the Firewall homepage

The configuration of the Firewall is stored in the active configuration of the ctrlX OS device. If the archive of a configuration is loaded using "Manage app data", the Firewall configuration overwrites the Firewall configuration currently active in the ctrlX OS device.

Elements of the expressions for rules

The possible elements of the rules are divided into the groups “Match packet payload”, “Match packet meta data”, “Match conntrack data”, “Destination NAT” or “Source NAT” and “Masquerade” as well as in a Counter element. Click on the  button “Add expression” to first select a group which elements should be available to create an expression.

A click on one of the element groups opens the editor for the expressions or a click on the “Counter” directly inserts a counter into the rule. The “Counter” element counts the number of packets at its inserted position. It also counts the total size of the packets in bytes run through the preceding expression.

The element selected in the field on the left of the expression editor changes the possible selection of elements on its right.

Element group	More fields to structure the rule - depending on the element group			
Match packet payload	Pro- tocol	Field	Operator	Input field for the value to be used on the value defined via the operator by the fields “Protocol” and “Field”
Match packet data	Key		Operator	Input/selection field to be used for the value on the value in the field “Key” via the operator
Match packet conn-track data	Key		Operator	Selection field to be used for the value on the value in the field “Key” via the operator
Destination NAT/ Source NAT	Input field for: IPv4 address to translate to/IPv6 address to translate to			Input field for: Port to translate to
Masquerade	Input field for: Port to translate to			
Counter	No further fields			

Elements of the first field in the element groups					
Match packet payload	Match packet meta data		Match conn-track data	Destination NAT/Source NAT	Masquerade
Protocol	Key		Key	IPv4 address to translate to*/IPv6 address to translate to*	Port
	Name	Description			
ip	iifname	Input interface name	state	Input field	Input field
ip6	iiftype	Input interface type	direction		
tcp	iif	Input interface index			
udp	oifname	Output interface Name			
ether	oiftype	Output interface type			
icmp	oif	Output interface index			
icmp6	length	Packet length			
	protocol	Packet protocol			
	pktype	Packet type			

More fields to structure the rule				
The availability of the elements depends on the first field: "Protocol" or "Key" or the IPv4 address to translate to*/IPv6 address to translate to*				
Field		Operator	Freely selectable input values. They depend on the value in the respective "Protocol" or "Key" field or the IpvIPv4/IPv6 address to translate to field	
Name	Description		Input value/selection value	Can be selected for the element group. The field name depends on the value in the field
saddr	source address	==	Free input	Match packet payload, "Protocol" and "Field" (for all values)
daddr	destination address	!=	ether	Match packet meta data, Key
version	IP Header version	<	ppp	Match packet meta data, Key
length	ip: total packet length ip6: payload length	>	ipip	Match packet meta data, Key
protocol	upper layer protocol	<=	ipip6	Match packet meta data, Key
nexthdr		>=	loopback	Match packet meta data, Key
sport	source port	in	sit	Match packet meta data, Key
dport	destination port		ipgre	Match packet meta data, Key
type			ip	Match packet meta data, Key
code			ip6	Match packet meta data, Key

More fields to structure the rule				
The availability of the elements depends on the first field: "Protocol" or "Key" or the IPv4 address to translate to*/IPv6 address to translate to*				
Field		Operator	Freely selectable input values. They depend on the value in the respective "Protocol" or "Key" field or the IpvIPv4/IPv6 address to translate to field	
Name	Description		Input value/ selection value	Can be selected for the element group. The field name depends on the value in the field
			arp	Match packet meta data, Key
			vlan	Match packet meta data, Key
			broadcast	Match packet meta data, Key
			unicast	Match packet meta data, Key
			multicast	Match packet meta data, Key
			Free input	Match packet meta data, Key (for values: iifname, iif, oif-name, oif, length)
			new	Match conntrack data, Key
			established	Match conntrack data, Key
			related	Match conntrack data, Key
			untracked	Match conntrack data, Key
			original	Match conntrack data, Key
			reply	Match conntrack data, Key
			Free input <i>port</i>	Destination NAT
			Free input <i>port</i>	Source NAT
			Free input <i>port</i>	Masquerade

Further information

- ➔ Chapter 5.2.1 tab – Filter on page 9
- ➔ Chapter 5.1.1 Window – Firewall on page 8
- ➔ Chapter 4.1 Firewall App on page 6
- ➔ Chapter 5.1.1 Window – Firewall on page 8
- ➔ ctrlX OS Runtime, Application Manual, chapter "Windows - Manage app data"

6 Related documentation

6.1 Overview

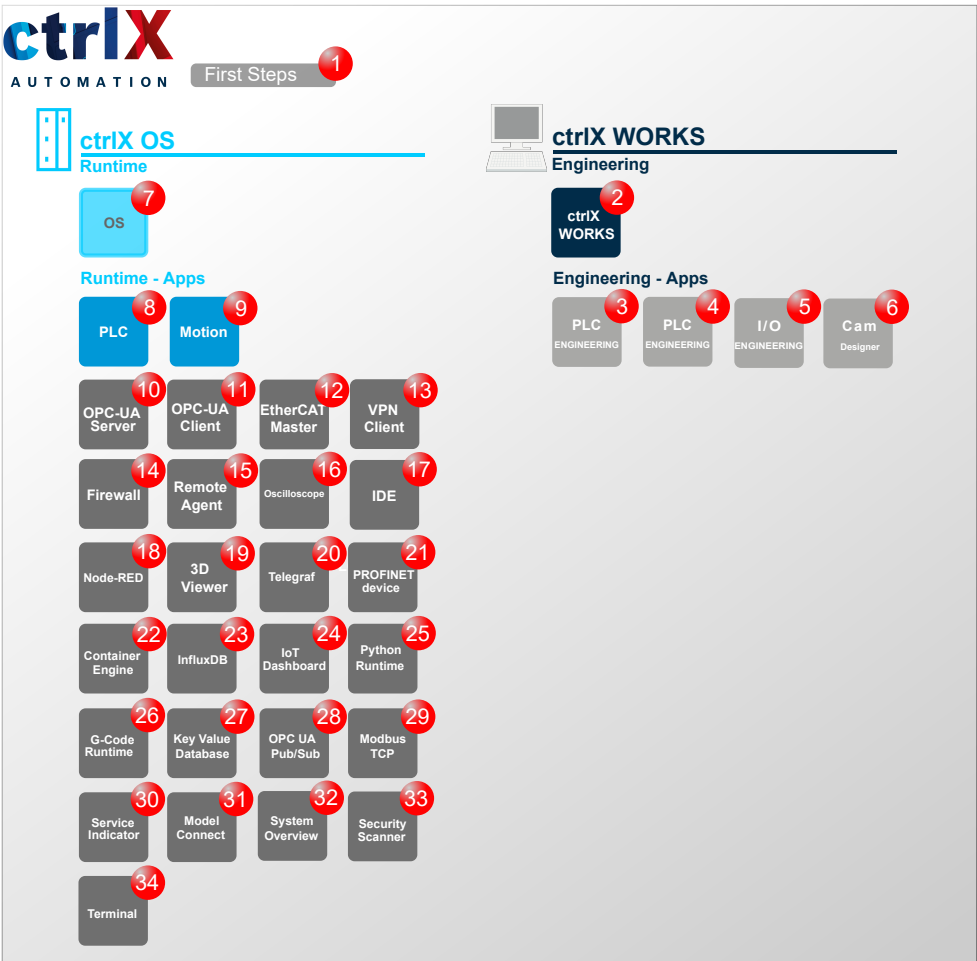


Fig. 10: Overview of the related documentation

6.2 ctrlX AUTOMATION

No.	Documentation
1	ctrlX WORKS - First Steps Quick Start Guide ↪ Link to the web documentation Ordering information: • DOK-XWORKS-F*STEP*****-QU01-EN-P • R911403760

6.3 ctrlX WORKS

No.	Documentation
2	ctrlX WORKS - Basic System Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XWORKS-WRK***V03**-APRS-EN-P • R911423376
3	ctrlX PLC Engineering - PLC Programming System Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XPLC**-PLE***V03**-APRS-EN-P • R911423378
4	ctrlX PLC Engineering - PLC Libraries Version 3 Reference ↗ Link to the web documentation Ordering information: • DOK-XPLC**-LIB***V03**-RERS-EN-P • R911423456
5	ctrlX I/O Engineering - Field Bus Configuration for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XIO***-IOE***V03**-APRS-EN-P • R911423380
6	ctrlX Cam Designer - Configuring the ctrlX MOTION Cams Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XWORKS-CAM***V03**-APRS-EN-P • R911427217

6.4 ctrlX OS

No.	Documentation
7	ctrlX OS - Operating System for ctrlX CORE Control Devices Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-XCR***V03**-APRS-EN-P • R911423382
	ctrlX OS - Nodes of the Data Layer Version 3 Reference ↗ Link to the web documentation Ordering information: • DOK-XCORE*-DL****V03**-RERS-EN-P • R911423384
	ctrlX OS - Diagnostics Version 3 Reference ↗ Link to the web documentation Ordering information: • DOK-XCORE*-DIAG**V03**-RERS-EN-P • R911423386

6.5 ctrlX OS Apps

No.	Documentation
8	PLC App - PLC Runtime Environment for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-PLC***V03**-APRS-EN-P • R911423401
9	Motion App - Motion Runtime Environment for ctrlX CORE Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-MOT***V03**-APRS-EN-P • R911423405
10	OPC UA Server App - OPC UA Server for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-UAS***V03**-APRS-EN-P • R911423392

No.	Documentation
11	OPC UA Client App - OPC UA Client for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-UAC***V03**-APRS-EN-P • R911423390
12	EtherCAT Master App - EtherCAT Master for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-ECM***V03**-APRS-EN-P • R911423394
13	VPN Client App - Remote Maintenance Software for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-VPN***V03**-APRS-EN-P • R911423388
14	Firewall App - Security Functions for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-FRW***V03**-APRS-EN-P • R911423397
15	Remote Agent App - ctrlX Device Portal Connection for ctrlX OS Devices Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-RMA***V03**-APRS-EN-P • R911423399
16	Oscilloscope App - Oscilloscope Function for ctrlX OS Devices Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-OSC***V03**-APRS-EN-P • R911423407
17	IDE App - Integrated Development Environment Version 2 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-IEN***V02**-APRS-EN-P • R911421612

No.	Documentation
18	Node-RED App - Graphical Programming for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-RED***V03**-APRS-EN-P • R911423403
19	3D Viewer App - Browser-based 3D Kinematics Simulation for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-3DV***V03**-APRS-EN-P • R911423411
20	Telegraf App - Server Agent for Collecting Data in the Data Layer Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-TSA***V03**-APRS-EN-P • R911425238
21	PROFINET Device App - PROFINET Device for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-PND***V03**-APRS-EN-P • R911425232
22	Container Engine App - Using Docker® Images on the ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-DOE***V03**-APRS-EN-P • R911425234
23	InfluxDB App - Influx Database Connection for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-IDB***V03**-APRS-EN-P • R911425240
24	IoT Dashboard App - Data Visualization in Dynamic, Interactive Dashboards Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-GDB***V03**-APRS-EN-P • R911425248

No.	Documentation
25	Python Runtime App - Python Runtime Environment for ctrlX CORE Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-PYR***V03**-APRS-EN-P • R911425244
26	G-Code Runtime App - G-Code Interpreter for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-GCO***V03**-APRS-EN-P • R911425246
27	Key Value Database App - Managing Data in the Data Layer Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-KVD***V03**-APRS-EN-P • R911425250
28	OPC UA Pub/Sub App - OPC UA Pub/Sub for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-UAP***V03**-APRS-EN-P • R911423409
29	Modbus TCP App - Modbus TCP Communication for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-MBT***V03**-APRS-EN-P • R911425236
30	Service Indicator App -Service Indicator for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-SIN***V03**-APRS-EN-P • R911425242
31	Model Connect App - Target for Model-Based Development and Simulation for ctrlX OS Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-MOC***V03**-APRS-EN-P • R911425252

No.	Documentation
32	System Overview App - System Topology and System Information Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-SOV***V03**-APRS-EN-P • R911425254
33	Security Scanner App - Inventory of Components Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-SSC***V03**-APRS-EN-P • R911427698
34	Terminal App - Terminal with Limited SSH Access Version 3 Application manual ↗ Link to the web documentation Ordering information: • DOK-XCORE*-TER***V03**-APRS-EN-P • R911428579

7 Service and support

Our worldwide service network provides an optimized and efficient support. Our experts provide you with advice and assistance. You can contact us **24/7**.

Service Germany

Our technology-oriented Competence Center in Lohr, Germany, is responsible for all your service-related queries for electric drive and controls.

Contact the **Service Hotline** and **Service Helpdesk** under:

Phone: **+49 9352 40 5060**

Fax: **+49 9352 18 4941**

Email: [↗ service.svc@boschrexroth.de](mailto:service.svc@boschrexroth.de)

Internet: [↗ http://www.boschrexroth.com](http://www.boschrexroth.com)

Additional information on service, repair (e.g. delivery addresses) and training can be found on our internet sites.

Service worldwide

Outside Germany, please contact your local service office first. For hotline numbers, refer to the sales office addresses on the internet.

Preparing information

To be able to help you more quickly and efficiently, please have the following information ready:

- Detailed description of malfunction and circumstances
- Type plate specifications of the affected products, in particular type codes and serial numbers
- Your contact data (phone and fax number as well as your e-mail address)

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